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Studierichting: Informatica
Oefeningenreeks 4A nr. 53.8

$$\int \sin^{-2} x \cos^{-4} x dx$$

$$= \int \frac{1}{\sin^2 x \cos^2 x} \cdot \frac{1}{\cos^2 x} dx$$

$$t = \tan x \Rightarrow dt = \frac{1}{\cos^2 x} dx$$

$$\Rightarrow \frac{1}{\cos^2 x} = 1 + \tan^2 x = 1 + t^2 \text{ en } \frac{1}{\sin^2 x} = 1 + \cot^2 x = 1 + \frac{1}{t^2}$$

$$= \int \frac{1}{(1+t^2)\left(\frac{t^2+1}{t^2}\right)} dt = \int \frac{t^2}{t^4 + 2t^2 + 1} dt$$

$$= \int (t^2 + 2 + t^{-2}) dt = \frac{1}{3} t^3 + 2t - \frac{1}{t} + C$$

$$= \frac{1}{3} \tan^3 x + 2 \tan x - \frac{1}{\tan x} + C$$

References